

UG10185

RW612 Matter-Zigbee Bridge User Guide

Rev. 3.0 — 11 December 2025

User guide

Document information

Information	Content
Keywords	UG10185, RW612, Matter, Zigbee, Matter-Zigbee bridge, OpenThread Border Router
Abstract	This document describes the NXP Matter-Zigbee bridge solution based on the RW612 MCU.



1 Introduction

Matter (formerly Project Connected Home over IP, or Project CHIP) is a new Working Group within the Connectivity Standards Alliance (CSA, formerly the Zigbee Alliance). This Working Group plans to develop and promote the adoption of a new, royalty-free connectivity standard to increase compatibility among smart home products, with security as a fundamental design tenet.

Traditional non-Matter devices cannot directly join the Matter network. However, Matter bridge solves the problem. A Matter bridge can join a Matter network as a Matter device, and a non-Matter device must be mapped to the Matter network as a dynamic endpoint. In this way, other Matter devices can communicate with a non-Matter device through its dynamic endpoint.

2 NXP Matter-Zigbee bridge solution

Currently, NXP offers two solutions for implementing a Matter-Zigbee bridge:

- One solution is based on the i.MX MPU platform.
- The other solution is based on the RW612 MCU.

This document focuses exclusively on the RW612 MCU-based solution.

The RW612 integrates Wi-Fi, Bluetooth Low Energy, and IEEE 802.15.4 within a single SoC. There are two variants of the MCU-based approach:

- RW612 as a Matter OpenThread Border Router (OTBR)
In this configuration, RW612 operates as a Matter OTBR with Wi-Fi and Thread interfaces. It connects via UART to a standalone K32W0x1 device acting as a Zigbee Coordinator. After RW612 is commissioned into a Matter network as a Matter over Wi-Fi node, OTBR is enabled on its 802.15.4 interface. Acting as an OpenThread leader, the OTBR allows additional Matter over Thread nodes to join the network.
- RW612 as a Matter over Wi-Fi device with integrated Zigbee Coordinator
In this variant, RW612 functions as a Matter over Wi-Fi node while simultaneously running a Zigbee Coordinator on the same device.

In both cases, the Matter-Zigbee bridge functionality is implemented on RW612. The device manages the Zigbee Coordinator to form a Zigbee network and permits other Zigbee nodes to join. Once a Zigbee node is commissioned, RW612 dynamically assigns an endpoint, enabling Zigbee devices to appear as native Matter nodes. These nodes can then be controlled by a Matter controller, such as the CHIP Tool (chip-tool).

2.1 Features

The following is a features list of the Matter-Zigbee bridge.

- Features specific to the RW612 + K32W0x1 variant:
 - Matter over Thread support
 - OpenThread border router (OTBR) functionality
 - Set up OTBR on RW612
 - Commission Matter over Thread devices into the Matter network
- Features common to both RW612-only and RW612 + K32W0x1 variants:
 - Matter over Wi-Fi support
 - Zigbee network management
 - Form and start a Zigbee network
 - Permit Zigbee nodes to join
 - Commission and remove Zigbee devices
 - List joined Zigbee nodes and retrieve their information after restart

- Remove joined Zigbee nodes
- Factory reset by erasing PDM
- Zigbee bridge functionality integrated into RW612
 - Seamless connection of Zigbee devices to the Matter ecosystem
 - Dynamic endpoint allocation for Zigbee nodes
- Supported Zigbee clusters
 - OnOff cluster
 - LevelControl cluster
 - ColorControl cluster
 - Temperature sensor cluster
- Matter CLI commands for configuration and control

2.2 Hardware requirements

The Matter-Zigbee bridge solution on RW612 is supported by both RW612 EVK A1 and FRDM-RW612 boards. However, the recommended platform is FRDM-RW612.

For the variant that includes OTBR support, an additional K32W0x1-DK6 kit (consisting of the MEZZANINE module and OM15076-3 Carrier Board) is required.

Required connections between FRDM-RW612 and K32W0x1-DK6:

- FRDM-RW612 J1 Pin 2 (GPIO_9) FC1_UART_RX ↔ K32W0x1-DK6 J3 Pin 15 USART0_TX
- FRDM-RW612 J1 Pin 4 (GPIO_8) FC1_UART_TX ↔ K32W0x1-DK6 J3 Pin 16 USART0_RX
- FRDM-RW612 J6 Pin 2 (GPIO_19) RST ↔ K32W0x1-DK6 J3 Pin 30 RSTN

Additional hardware:

- Matter controller (compatible with Matter 1.4 or newer)
- USB-UART converter for Matter log output
 - FRDM-RW612 J5 Pin 4 (GPIO_03) FC0_UART_TXD ↔ Converter RX
 - FRDM-RW612 J5 Pin 8 (GND) ↔ Converter GND
- K32W0x1 as Zigbee ColorLight
- K32W0x1 as Zigbee LightTemperatureOccupancy (LTO) sensor

For variant with OTBR support:

- FRDM-MCXW71 as Matter over Thread Light, to exercise Matter over Thread device joining to the Thread network created by the OTBR

2.3 Software requirements

The following list describes the software requirements for the Matter Zigbee Bridge setup.

- Software requirements common to both variants:
 - Ubuntu 22.04 as a standalone PC or installed in a Virtual Machine, such as VirtualBox
 - J-Link version (> v.792f)
 - Embedded software development tools, such as CMake, Ninja, and GCC for Arm
 - Software versioning tools, such as Git
 - Zigbee Color Light: *JN-AN-1244\Binaries\ExtendedColorLight_GpProxy_OM15081\ExtendedColorLight_GpProxy_OM15081.bin (Prebuild)*
 - Zigbee Temperature Sensor: *JN-AN-1246\Binaries\LTOSensor_NtagIcode_Ota_OM15081R2\LTOSensor_NtagIcode_Ota_OM15081R2_V1.bin (Prebuild)*

- RW612 Matter Zigbee Bridge: binary available in the build folder according to the configuration
- For the OTBR variant, additional software is required:
 - K32W0x1 SDK 2.6.16
 - DK6Programmer.exe: SDK_2_6_16_K32W061DK6\tools\ JN-SW-4407-DK6-Flash-Programmer\JN-SW-4407DK6 Production Flash Programmer v4564.exe
 - Zigbee control bridge: JN-AN-1247\Binaries\ControlBridge_Full_GpProxy_115200\ControlBridge_Full_GpProxy_115200.bin (build with baud rate set to 115200 bit/s)
 - Matter over Thread Light: see [Matter Lighting App Example](#) for details on how to build Matter application

3 Build Matter-Zigbee bridge application

To build a Matter-Zigbee bridge application, run the following commands depending on the used configuration (with or without OTBR support).

For the configuration with OTBR support:

1. Create the build environment:

```
git clone -b rw612_with_external_transceiver
https://github.com/nxp-appcodehub/dm-matter-zigbee-bridge-rw612.git
```

2. Initialize the Matter repo as submodule inside `bridge-app/nxp/rt/rw61x/third_party`:

```
cd dm-matter-zigbee-bridge-rw612
git submodule update --init
```

3. Copy the patches from patches folder to `bridge-app/nxp/rt/rw61x/third_party`:

```
cp patches/* bridge-app/nxp/rt/rw61x/third_party/
```

4. Copy the `zigbee_bridge` folder to `bridge-app/nxp/rt/rw61x/third_party/matter/third_party/nxp`:

```
cp -r zigbee_bridge/ bridge-app/nxp/rt/rw61x/third_party/matter/third_party/
nxp/
```

5. Go to the matter root folder:

```
cd bridge-app/nxp/rt/rw61x/third_party/matter/
```

6. Install the required packages:

```
sudo apt-get install git gcc g++ pkg-config libssl-dev libdbus-1-dev
libglib2.0-dev
libavahi-client-dev ninja-build python3-venv python3-dev python3-pip unzip
libgirepository1.0-dev libcairo2-dev libreadline-dev
```

7. Checkout NXP-specific Matter submodules:

```
scripts/checkout_submodules.py --shallow --platform nxp --recursive
```

8. Run the bootstrap script to initialize the Matter build environment:

```
source scripts/bootstrap.sh
```

9. If bootstrap is already done, the environment activation can be done by using the `activate` script:

```
source scripts/activate.sh
```

Note: The environment is active only in the terminal in which the `bootstrap/activate` script was executed.

10. Apply the following patches for the `/src` and `/example` folders:

```
git apply ../example-changes.patch
git apply ../src-changes.patch
```

11. Initialize the NXP SDK:

```
third_party/nxp/nxp_matter_support/scripts/update_nxp_sdk.py --platform
common
```

12. Go the `nxp_matter_support` folder:

```
cd third_party/nxp/nxp_matter_support/
```

13. Apply the following patch to modify the board files and enable the UART communication with the Zigbee coordinator:

```
git apply ../../../../sdk-changes.patch
```

14. Build the FRDM-RW612 Matter-Zigbee bridge app:

- a. Go back to the `bridge-app/nxp/rt/rw61x` folder:

```
cd ../../../../
```

- b. Generate the build files for the FRDM-RW612 board and with the Thread BR option:

```
gn gen --args="chip_enable_wifi=true chip_enable_openthread=true nxp_enable_matter_cli=true  
board_version=\"frdm\" out/debug
```

- c. Build the example:

```
ninja -C out/debug/
```

After the build finishes, the application binary (`.srec` file) can be found in the `out/debug/` folder under the name `chip-rw61x-bridge-example.srec`.

15. Flash the application binary on the board by following these instructions:

```
https://github.com/NXP/ot-nxp/tree/release/v1.4.0/src/rw/rw612#flash-binaries
```

For the configuration without OTBR support:

1. Create the build environment and set up Matter environment:

```
git clone -b rw612_only https://github.com/nxp-appcodehub/dm-matter-zigbee-  
bridge-rw612.git
```

All the information required to set up the environment, build the application, and test, are available in the Matter Documentation for NXP MCU platforms:

```
https://docs.mcuxpresso.nxp.com/matter/latest/html/index.html
```

To set up the environment, the local `west.yml` should be used to clone all the repos (`west init -l [--mf FILE] directory` OR `west init -m https://github.com/nxp-appcodehub/dm-matter-zigbee-bridge-rw612.git sdk-next --mf west.yml --mr rw612_only`)

2. Build the FRDM-RW612 Matter-Zigbee bridge app:

- a. Go back to the `dm-matter-zigbee-bridge-rw612` folder

```
cd ../../../../dm-matter-zigbee-bridge-rw612
```

- b. Build the example:

```
west build -b frdmrw612 bridge-app/nxp/
```

Note: If the SDK is placed in a specific folder, the path to Matter in `CMakeList.txt` (`bridge-app/nxp/CMakeLists.txt:25 -> ${CHIP_ROOT}`) must be modified.

3. Flash the application binary on the board by following these instructions:

```
https://github.com/NXP/ot-nxp/tree/release/v1.4.0/src/k32w0/k32w061#flash-binaries
```

4 Matter-Zigbee bridge demo setup

This section describes the setup and topology of the Matter-Zigbee bridge.

4.1 Matter-Zigbee bridge topology

Figure 1 shows that the wireless MCU RW612 has integrated tri-radio. Bluetooth Low Energy and Wi-Fi are used in sequence during its commissioning to the Matter controller.

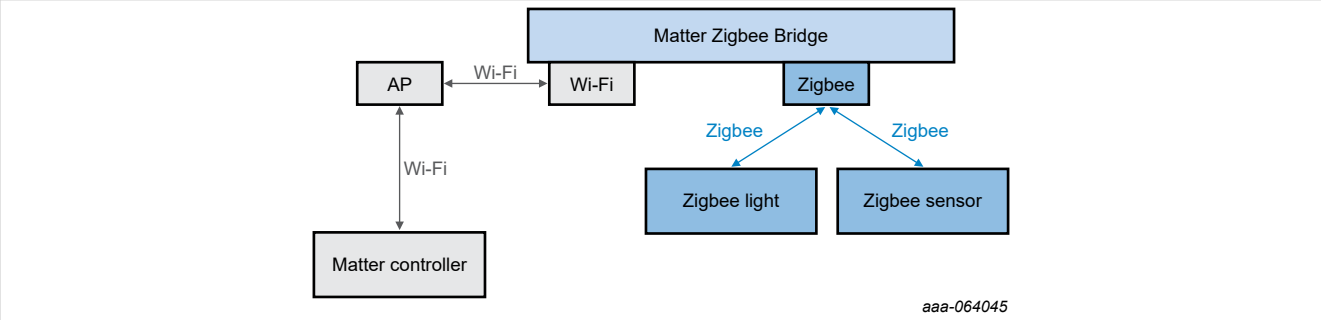


Figure 1. RW612 Matter-Zigbee bridge topology-Matter over Wi-Fi configuration

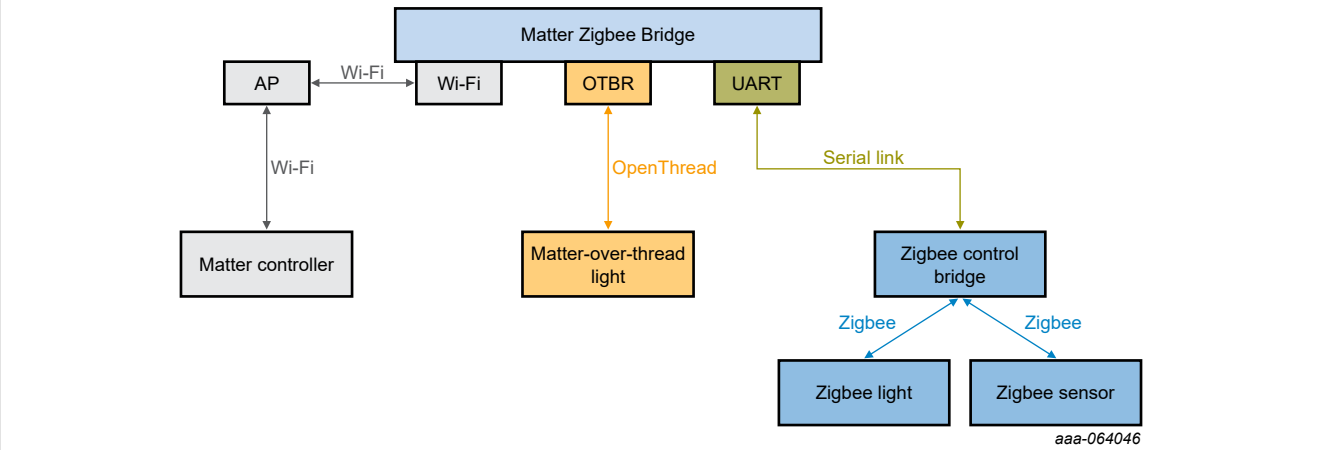
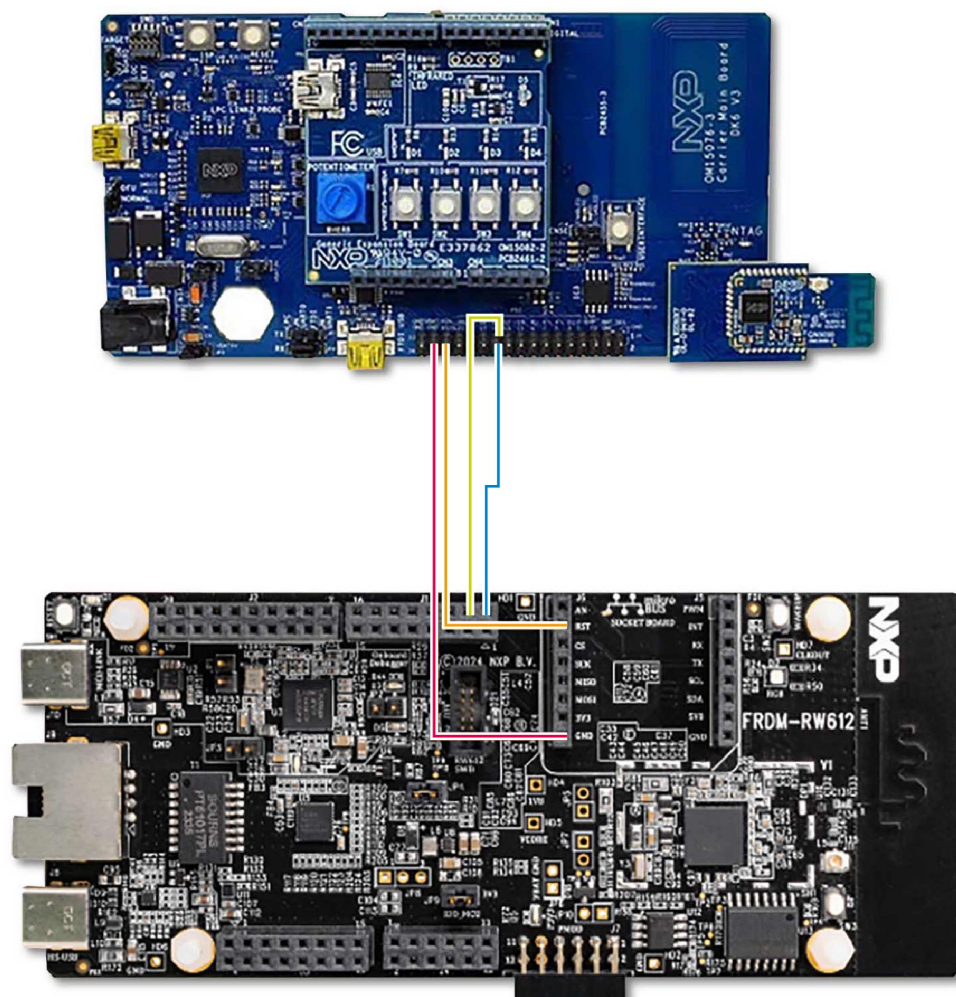


Figure 2. RW612 Matter-Zigbee bridge topology-OTBR configuration

OpenThread runs over 802.15.4 and can implement OTBR role as a leader, enabling other Matter over Thread devices to be commissioned to the Matter network through this OTBR.

For the configuration with OTBR support, an additional standalone K32W0x1 Zigbee Coordinator can be controlled by RW612 via the UART interface, using its Matter-Zigbee bridge functionality. This forms a Zigbee network that permits any Zigbee devices to join it with dynamic endpoints allocated by RW612. As a result, all Zigbee devices are considered as Native Matter nodes and can be controlled using chip-tool commands on the Matter controller via these dynamic endpoints.



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Figure 3. Matter-Zigbee bridge hardware platform

Figure 3 shows the connection between RW612 and K32W0x1. In addition to the Txd/Rxd signals, the UART interface includes three signals: VCC, GND, and Reset.

RW612 is combined with MatterOverWifi + OTBR + Matter Zigbee Bridge while K32W0x1-DK6 as Zigbee Coordinator is built from NXP reference code (see [AN1247](#)) with an exception that UART baud rate has been changed from default 1000000 bit/s to 115200 bit/s for the sake of communication stability.

4.2 Matter-Zigbee bridge setup

This section describes the Matter-Zigbee bridge setup process in detail.

4.2.1 Join RW612 to Matter network via Wi-Fi

Run the following command on Matter controller after RW612 boot-up with the message “CHIPoBLE advertising started” in the end of logs on its UART console.

```
$ chip-tool pairing ble-wifi 1 SSID Passwd 20202021 3840
```

Wait until the message “Device commissioning completed with success” is visible at the end of logs on the Matter controller console. This message confirms that RW612 has successfully joined the Matter network as a MatterOverWifi device.

4.2.2 Set up Matter-Zigbee bridge

To form a Zigbee network on the K32W0x1-DK6 board module, run the following commands on the RW612 CLI. The module connects to RW612 through its UART interface and this permits other Zigbee nodes to join it.

```
>zb-erasepdm      --- erase currently used Zigbee channel, skip this if want to continue on
                    existing channel
>zb-nwk-form 11    --- can be any value between 11~26 as valid Zigbee channel
>zb-nwk-pjoin 255  --- 255 to enable and 0 to disable permit join
```

Power on factory-new color light K32W0x1, the following messages appear on the RW612 console:

```
Add Color Light
Node Type=3,Short=0x18d4,MAC=0x158d00031f1742,EP=12928
```

Where EP=12928 is the dynamic allocated endpoint that maps the Non-Matter Zigbee node as a Matter device. If more Zigbee nodes join the RW612 Matter network, then increased dynamic endpoint as 12929, 12930, and so on, are assigned to these additional nodes individually.

4.2.3 Control Zigbee nodes through dynamic endpoint by Chip-Tool on Matter controller

Red, Green, Blue (RGB) on Color Light of the K32W0x1 device can be changed according to each of the following commands:

```
$ chip-tool onoff toggle 1 12928
$ chip-tool levelcontrol move-to-level 2 1 1 1 1 12928
$ chip-tool colorcontrol move-to-hue 32 1 1 1 1 12928
$ chip-tool colorcontrol move-to-saturation 64 1 1 1 1 12928
$ chip-tool colorcontrol move-to-color-temperature 128 1 1 1 1 12928
$ chip-tool colorcontrol move-to-color 30000 60000 1 1 1 1 12928
```

Note: The description of each argument after cluster and its corresponding command can be referred by running “chip-tool cluster command”. For example:

```
chip-tool colorcontrol move-to-hue Hue Direction TransitionTime OptionsMask
OptionsOverride destination-id endpoint-id-ignored-for-group-commands
```

in the response of the above command “Usage”, all supported parameters are listed.

4.2.4 Set up OTBR on RW612

Run the following commands one by one on the RW612 CLI:

```
>otcli dataset init new
>otcli dataset panid 0xabcd --- 0xabcd can be changed to other value
>otcli dataset channel 25 --- 25 can be changed between 11~26
>otcli dataset commit active
>otcli ifconfig up
>otcli thread start
>otcli state --- must wait till “leader” state appears
>otcli dataset active -x --- thread dataset used in chip-tool similar to
following: 0e08000000000001000035060004001fffe002088711152e77458a490708fdcbf744a9102
```

```
0cb05100c208752e1bd2586f0a87ed481890312030f4f70656e5468726561642d633130640410d60d95cb5db  
1044086f7813e66de19020c0402a0f7f80102abcd0003000019
```

4.2.5 Join MCX W71 Matter over Thread light app to RW612 OTBR

To see the message “Started BLE Advertising” on its UART console, press SW2 on MCX W71. Afterward, run the following command on the Matter controller console:

```
$chip-tool pairing ble-thread 2 hex:  
0e08000000000001000035060004001fffe002088711152e77458a490708fdcbf744a91020cb05100  
c208752e1bd2586f0a87ed481890312030f4f70656e5468726561642d633130640410d60d95cb5db1044086f7813e66de  
19020c0402a0f7f80102abcd0003000019 20202021 3840
```

Wait until the message “Device commissioning completed with success” is visible at the end of logs on the Matter controller console. The message confirms that MCX W71 has successfully joined the Matter network through RW612 OTBR as a Matter over Thread device.

To control On/Off of the MCX W71 LED, use the following command:

```
$chip-tool onoff toggle 2 1
```

5 Acronyms

Table 1. Acronyms

Term	Definition
CHIP	Connected Home over IP
CLI	Command-line interface
EVK	Evaluation kit
IDE	Integrated development environment
IP	Internet protocol
OTBR	OpenThread Border Router
SDK	Software development kit
SW	Software

6 References

- *Webpage - SEGGER J-Link tool* ([link](#))
- *Application Note - Zigbee 3.0 ControlBridge UserGuide* (document [AN1247](#))

7 Note about the source code in the document

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8 Revision history

[Table 2](#) summarizes revisions to this document.

Table 2. Revision history

Document ID	Release date	Description
UG10185 v.3.0	11 December 2025	Updated Section 2 , Section 2.1 , Section 2.2 , Section 2.3 , Section 3 , Section 4.1 , Section 4.2
UG10185 v.2.0	17 February 2025	- Updated Section 3 - Updated Section 6
UG10185 v.1.0	23 December 2024	Initial public release

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